

Let N be a prime number > 3 . Mazur has defined, from the theory of modular forms, a unit u in $\mathbb{Z}/N$. This unit turned out to be, up to a 6th root of unity, $\prod_{k=1}^{(N-1)/2} k^k$. In this talk we will describe how the unit is connected to various objects in number theory. For instance:

- The unit u can be understood as a derivative of the zeta function at -1 , (despite living in a finite field).

- Lecouturier has shown that this unit is the discriminant of the Hasse polynomial: $\sum_{i=0}^{(N-1)/2} a_i X^i$ modulo N , where a_i is the square of the i -th binomial coefficient in degree N .

- Calegari and Emerton have related u to the class group of the quadratic field $\mathbb{Q}(\sqrt{-N})$. For every prime number p dividing $N - 1$, It is important to determine when u is a p -th power in $(\mathbb{Z}/N)^*$. If time allows, I will describe the connections to modular forms and Galois representations, and the general theory that Lecouturier has developed from this unit. For instance, when u is not a p th power, a certain Hecke algebra acting on modular forms is of rank 1 over the ring of p -adic integers $\mathbb{Z}_p$ (the original motivation of Mazur). The unit plays an important role in the developments around the conjecture of Harris and Venkatesh.